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St. Joseph's college f Arts & Science (Autonomous)

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B. Sc DEGREE EXAMINATION-APRIL 2021

EBC613- ENDOCRINOLOGY

Time: 3 hrs **Max Marks: 75**

SECTION-A (25X1=25)

Answer ALL the questions

I. Choose the best answer: (10X1=10)

1. The following has been referred to as the master endocrine gland of vertebrates

   a) Thyroid b) Pituitary c) Pancreas d) none of these

2. _____ is the maintenance of a constant internal environment by the organism

   a) Saturation b) Internalisation

   c)Homeostasis d)all the above

3. Protein kinase C is a _____

   a) Ser/Thr kinase b) Tyr/Thr kinase

   c) Tyr kinase d)none of the above

4. The Ras-MAP kinase pathway is activated by _____

   a) peptide hormones b) growth factors

   c) steroid hormones d) all the above

5. _____ hormones are also known as tropic hormones

   a) Thyroid b) Hypothalamic c) Pituitary d) Gonadal

6. Anterior and posterior pituitary are connected by _____

   a) parstuberalis b) pars intermedia

   c) pars nervosa d) pars distalis

7. _____ is the major component of colloid inside the thyroid follicle

   a) Thyroglobulin b) MIT and DIT

   c) T3 and T4 d) Free iodine

8. Which of the following stimulates calcium absorption from the intestine?

   a) calcitonin b) calmodulin

   c) vitamin D d) none of the above

9. Active form of androgens is _____

   a) progesterone b) testosterone

   c) dihydrotestosterone d) androstenedione

10. The plasma testosterone level of normal men is _____

    a) 0.7 mg/dl b) 1.3 mg/dl c) 0.3 mg/dl d) 1.07 mg/dl

II. Say true or false (5X1=5)

11. Group II hormones are lipophilic in nature
12. Paracrine signalling helps in synaptic transmission
13. Posterior pituitary gland is also called as adenohypophysis
14. TSH is also known as thyrotropin
15. Platelet aggregation is inhibited by catecholamines

III. Answer all the questions (10X1=10)

16. Define endocrine system
17. Why hormonal signalling is slower than neuronal signalling?
18. Define diabetes insipidus
19. What is TSI?

20. Mention the biological role of dopamine
21. Give one function of vasopressin.
22. What is goiter?
23. Give the medullary hormones.
24. What is Addison's disease?
25. Give one function of parathyroid hormones.

SECTION-B (4X5=20)
Answer any FOUR of the following:

26. What are hormones? How they are classified?
27. Describe the mechanism of cAMP mediated signal transduction
28. Discuss the biological importance of posterior pituitary hormones
29. Write a short note on causes, symptoms and treatment methods of Grave's disease
30. Explain the structure and functions of insulin
31. Explain the biosynthesis and metabolic functions of catecholamines

SECTION-C (3X10=30)
Answer any THREE of the following:

32. Elaborate the various endocrine glands and their hormones with a neat diagram
33. Explain the mechanism of signal transduction through receptor
34. tyrosine kinases
35. Give a detailed account on structure and functions of hypothalamic hormones
36. Explain the synthesis, secretion, structure and functions of thyroid hormones
37. Give an elaborate note on biosynthesis and biological functions of androgens

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